P-005 | Fall 2026 cohort

Mathematics for Climate Science

Utrecht University

Your interests: **I don't really know. I enjoy maths and solving puzzles, but I also care a lot about climate change and would like to do something useful.**

Reference programme: Wiskunde en Toepassingen, Utrecht University ↗

Mathematical foundations and problem solving

- Calculus & Linear Algebra I — 7.5 EC
- Proofs in Mathematics — 7.5 EC
- Calculus & Linear Algebra II — 7.5 EC
- Analysis — 7.5 EC
- Probability — 7.5 EC
- Introduction to Groups & Rings — 7.5 EC
- Introduction to Multivariable Analysis — 7.5 EC
- Complex Analysis — 7.5 EC
- Differential Equations — 7.5 EC
- Statistics — 7.5 EC

Computing, modelling and AI

- Programming for Mathematics — 7.5 EC
- Methods & Models in Complex Systems — 7.5 EC
- Numerical Methods for Physicists & Astronomers — 7.5 EC
- Modelling with ODEs & PDEs — 7.5 EC
- Introduction to Machine Learning — 7.5 EC
- Introduction to Nonlinear Dynamical Systems — 7.5 EC
- Introduction to Scientific Computing — 7.5 EC

Climate science, Earth systems and solutions

- Atmosphere–Ocean Dynamics — 7.5 EC
- Fluid Mechanics & Transport Phenomena — 7.5 EC
- Climate Dynamics — 7.5 EC
- Geophysical Fluid Dynamics — 7.5 EC
- Turbulence in Fluids — 7.5 EC

Research, projects and practical application

- Bachelor Thesis — 15 EC

Current official example study path within Wiskunde en Toepassingen.

UCR — Mathematics, Models & Climate

greatest disciplinary depth

Your interests: **I don't really know. I enjoy maths and solving puzzles, but I also care a lot about climate change and would like to do something useful.**

Mathematical foundations and problem solving

- Mathematical Methods
- Logic & Discrete Mathematics
- Linear Algebra
- Advanced Calculus
- Probability & Statistics
- Linear Systems

Computing, modelling and AI

- Introduction to Programming
- Introduction to Data Science
- Computational & Numerical Methods
- Machine Learning
- Algorithms & Data Structures
- Image Processing & Computer Vision
- Robotics
- Optimization Methods in AI
- Neural Networks & Deep Learning

Climate science, Earth systems and solutions

- Sustainability Essentials I
- Sustainability Essentials II
- Climate Change
- Dynamic Earth
- Earth Observation
- The Science of Evolution
- Earth System Science Tools

Governance, economics and ethics

- AI Ethics

Academic development

- Personal & Professional Development

UCR — Quantitative Climate Action

more balanced across your interests

Your interests: **I don't really know. I enjoy maths and solving puzzles, but I also care a lot about climate change and would like to do something useful.**

Mathematical foundations and problem solving

- Mathematical Methods
- Linear Algebra
- Probability & Statistics

Computing, modelling and AI

- Introduction to Programming
- Introduction to Data Science
- Computational & Numerical Methods
- Machine Learning
- Algorithms & Data Structures
- Image Processing & Computer Vision
- Optimization Methods in AI
- Neural Networks & Deep Learning

Climate science, Earth systems and solutions

- Sustainability Essentials I
- Sustainability Essentials II
- Climate Change
- Dynamic Earth
- Earth Observation
- Earth System Science Tools
- Renewable System Design

Research, projects and practical application

- Research in a Sustainable Delta
- Consumer Product Design

Governance, economics and ethics

- Ethics
- Governance of Sustainability
- Peace & Conflict

Academic development

- Personal & Professional Development

UCR — Climate, Systems & Society

broader themes connecting your interests

Your interests: **I don't really know. I enjoy maths and solving puzzles, but I also care a lot about climate change and would like to do something useful.**

Mathematical foundations and problem solving

- Mathematical Methods
- Probability & Statistics
- Research Methods & Statistics

Computing, modelling and AI

- Introduction to Programming

Climate science, Earth systems and solutions

- Sustainability Essentials I
- Sustainability Essentials II
- Climate Change
- Earth Observation
- Earth System Science Tools
- Renewable System Design
- Life Cycle Analysis & Circular Economy
- Environmental Biogeochemistry
- Earth's Resources

Research, projects and practical application

- Research in a Sustainable Delta

Governance, economics and ethics

- Ethics
- Governance of Sustainability
- Introduction to Economics
- Law, Society & Justice
- Foundations of Comparative Politics
- Business & Economy in the Age of AI
- Introduction to Public International Law
- European Union Politics
- Behavioral Economics

Academic development

- Personal & Professional Development